

14-Service-Technical-Specifications.md

Healthcare Clinical Intelligence Platform

Service-by-Service Technical Specifications

Version: 1.0

Status: Engineering Ready

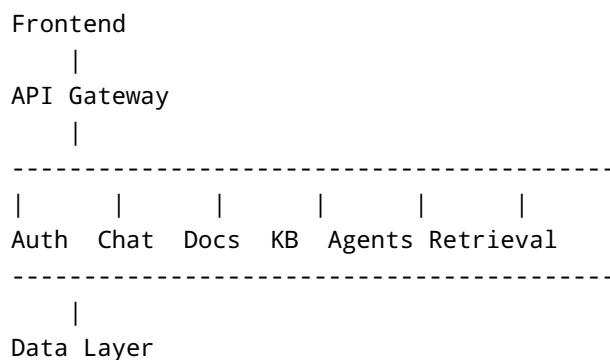
Purpose

This document translates architecture into implementation.

Every service must define:

- Responsibilities
 - Inputs
 - Outputs
 - APIs
 - Database Access
 - Dependencies
 - Observability
 - Failure Handling
-

Service Architecture



Service 1

Auth Service

Responsibilities

Authentication

Authorization

JWT Validation

Session Management

RBAC

Dependencies

Supabase

Redis

APIs

POST /auth/login

POST /auth/logout

POST /auth/refresh

GET /auth/me

Observability

Login Attempts

Failed Logins

Session Count

Service 2

Organization Service

Responsibilities

Organizations

Departments

Users

Roles

Permissions

APIs

GET /organizations

POST /organizations

GET /departments

POST /departments

Service 3

Document Service

Responsibilities

Upload

Versioning

Approval Workflow

Metadata Management

APIs

POST /documents

GET /documents

GET /documents/{id}

DELETE /documents/{id}

Storage

S3

Supabase

Service 4

Ingestion Orchestrator

Responsibilities

Coordinate ingestion workflow.

Workflow

```
graph TD;
  Upload --> Parse;
  Parse --> Metadata;
  Metadata --> Chunk;
  Chunk --> Embed;
  Embed --> Index;
```

Queue

Redis + Celery

Service 5

Parsing Service

Tool

Docling

Responsibilities

Extract:

Text

Tables

Images

Layout

Metadata

Outputs

Parsed Document JSON

Service 6

OCR Service

Tool

PaddleOCR

Responsibilities

Scanned PDFs

Images

Forms

Service 7

Metadata Enrichment Service

Responsibilities

Extract:

Medical Specialty

Disease

Drug

Procedure

Risk Level

Audience

Outputs

Enriched Metadata

Service 8

Ontology Service

Responsibilities

Map Entities To

SNOMED CT

RxNorm

ICD-10

LOINC

Outputs

Normalized Medical Entities

Service 9

Chunking Service

Responsibilities

Adaptive Chunking

Hierarchical Chunking

Parent Child Chunking

Graph-Aware Chunking

Chunkers

ResearchPaperChunker

GuidelineChunker

SOPChunker

TableChunker

SemanticChunker

Outputs

Chunk Objects

Service 10

Embedding Service

Responsibilities

Text Embeddings

Image Embeddings

Table Embeddings

Models

BGE-M3

ColQwen

Service 11

Graph Extraction Service

Responsibilities

Entity Extraction

Relationship Extraction

Knowledge Graph Building

Storage

Neo4j

Service 12

Indexing Service

Responsibilities

Store Data

Qdrant

Neo4j

Supabase

Validation

Before Indexing

Service 13

Retrieval Service

Responsibilities

Dense Retrieval

Sparse Retrieval

Metadata Retrieval

Parent Child Retrieval

Sources

Qdrant

Elasticsearch

Neo4j

Service 14

Query Rewriting Service

Responsibilities

Query Expansion

Multi Query Generation

Ontology Expansion

Outputs

Optimized Queries

Service 15

Reranking Service

Responsibilities

Rerank Results

Remove Noise

Prioritize Evidence

Models

BGE Reranker

Cohere Rerank

Service 16

Confidence Engine

Responsibilities

Calculate

Evidence Confidence

Source Confidence

Retrieval Confidence

Final Confidence

Output

```
{  
  "confidence":0.92  
}
```

Service 17

Corrective Retrieval Service

Responsibilities

Retry Retrieval

Rewrite Queries

Expand Search

Escalate Sources

Max Retries

3

Service 18

Planner Agent Service

Responsibilities

Intent Detection

Risk Classification

Retrieval Strategy

Tool Selection

Service 19

Retriever Agent Service

Responsibilities

Coordinate Retrieval

Evidence Collection

Service 20

Validator Agent Service

Responsibilities

Evidence Validation

Source Validation

Conflict Detection

Service 21

Safety Agent Service

Responsibilities

Emergency Detection

Risk Detection

Medical Safety

Authority

Can block responses.

Service 22

Response Agent Service

Responsibilities

Generate Responses

Generate Citations

Generate Confidence Explanations

Service 23

Memory Service

Responsibilities

User Memory

Department Memory

Organization Memory

Agent Memory

Storage

Postgres

Qdrant

Redis

Service 24

Evaluation Service

Responsibilities

RAGAS

DeepEval

Golden Dataset

Regression Testing

Service 25

Feedback Service

Responsibilities

Capture User Feedback

Store Ratings

Generate Improvement Signals

Service 26

Audit Service

Responsibilities

Track Everything

Events

Login

Query

Upload

Approval

Deletion

Memory Changes

Service 27

Notification Service

Responsibilities

Approval Requests

System Alerts

Security Alerts

Failures

Service 28

Observability Service

Responsibilities

Metrics

Logs

Traces

Cost Tracking

Tools

Langfuse

Prometheus

Grafana

OpenTelemetry

Service 29

API Gateway

Responsibilities

Authentication

Rate Limiting

Request Validation

Tenant Isolation

Logging

Service 30

Search Federation Service

Responsibilities

Combine Results From

Qdrant

Neo4j

Elasticsearch

External Sources

Outputs

Unified Evidence Package

Internal Event Bus

Events

```
document_uploaded
document_parsed
metadata_generated
chunking_completed
embedding_completed
graph_updated
retrieval_completed
evaluation_completed
feedback_received
```

Service Ownership

Core Platform

Auth

Organizations

Documents

AI Platform

Retrieval

Agents

Memory

Evaluation

Infrastructure

Observability

Deployment

Monitoring

Engineering Principles

Single Responsibility

Event Driven

Observable

Retryable

Governed

Multi-Tenant

Cost Aware

Final Principle

Every service should be independently deployable, independently observable, independently scalable, and independently testable.

This architecture should allow future migration to microservices without major rewrites.